Gebze Technical University

Department of Mechanical Engineering

Recitation 5

Harmonic Force Response

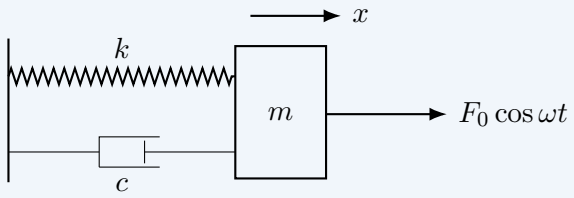
Lecturer: Saeed Lotfan | saeedlotfan.com

Self-study guide. These 30 questions develop the Week 5 topics through explanations and worked calculations. Each question is followed immediately by its solution.

Levels: Foundation builds the basic relations; Application uses them in a specified model; Challenge combines several steps or requires closer interpretation. Angles are in radians unless degrees are stated. Displacements are measured from static equilibrium. Amplitudes are nonnegative; phase carries the sign information.

Question 1 | Equation terms under harmonic forcing — Foundation

A fixed-base mass–spring–damper is driven by $F_0 \cos \omega t$. Write its forced-motion equation. For the steady response $x = X \cos(\omega t - \phi)$, identify the amplitude and relative phase of each term on the left-hand side. Explain why the force frequency is not generally the natural frequency.



Solution 1

1. Balance the applied and resisting forces

$m\ddot{x} = F_0 \cos \omega t - c\dot{x} - kx$, hence

$$\boxed{m\ddot{x} + c\dot{x} + kx = F_0 \cos \omega t.}$$

2. Differentiate the steady motion

Let $\psi = \omega t - \phi$. Then $\dot{x} = -\omega X \sin \psi$ and $\ddot{x} = -\omega^2 X \cos \psi$. Thus kx has amplitude kX in phase with x; $c\dot{x}$ has amplitude $c\omega X$ and leads x by 90°; $m\ddot{x}$ has amplitude $m\omega^2 X$ and is opposite to x. These are equation-term vectors, not the spring and damper forces on a free-body diagram, which have negative signs.

3. Distinguish the frequencies

ω is prescribed by the forcing mechanism; $\omega_n = \sqrt{k/m}$ is a property of the model. The steady forced response oscillates at ω. A homogeneous part may also be present while the initial-condition response persists.

Question 2 | Undamped response with arbitrary initial conditions **Foundation**

For $m\ddot{x} + kx = F_0 \cos \omega t$, $\omega \neq \omega_n$, derive the complete response with $x(0) = x_0$ and $\dot{x}(0) = v_0$. Explain the sign of the harmonic coefficient above resonance.

Solution 2

1. Find a particular solution

Try $x_p = P \cos \omega t$. Substitution gives $(k - m\omega^2)P = F_0$, so $P = F_0/(k - m\omega^2)$.

2. Add the free part and impose both conditions

Write $x = C \cos \omega_n t + D \sin \omega_n t + P \cos \omega t$. At $t = 0$, $C + P = x_0$ and $D\omega_n = v_0$. Therefore

$$\boxed{x = (x_0 - P) \cos \omega_n t + \frac{v_0}{\omega_n} \sin \omega_n t + P \cos \omega t.}$$

3. Interpret the sign and the limit

When $\omega > \omega_n$, $P < 0$. The amplitude is $|P|$ and the forced displacement is 180° out of phase with the force. Undamped homogeneous motion never decays, so the total response need not become x_p. At exact resonance the assumed trial function fails; it must be replaced by a term proportional to $t \sin \omega_n t$.

Question 3 | Below-resonance response — Application

An undamped system has $m = 10$ kg, $k = 4000$ N/m, and $F(t) = 400\cos(10t)$ N. Find the total response for (a) $x_0 = 0.1$ m, $v_0 = 0$; (b) $x_0 = 0$, $v_0 = 10$ m/s; (c) $x_0 = 0.1$ m, $v_0 = 10$ m/s. Interpret the plotted response.

Solution 3

1. Compute the frequency and signed forced coefficient

$$\omega_n = 20 \text{ rad/s}, \qquad P = \frac{400}{4000 - 10(10)^2} = 0.13333333 \text{ m}.$$

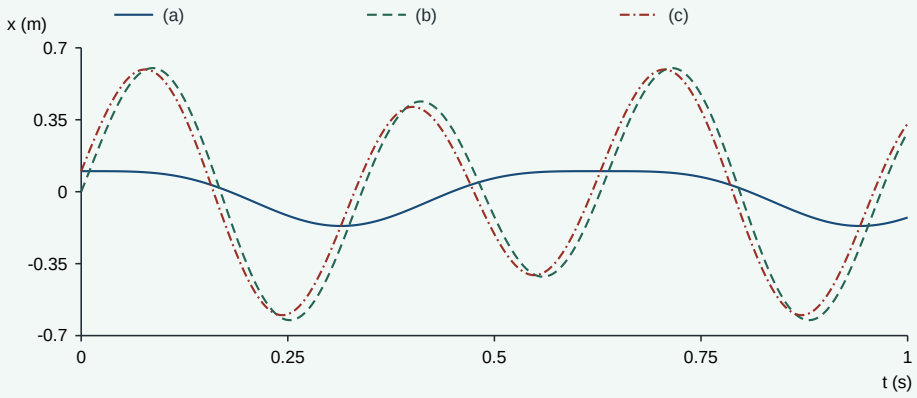
2. Apply the initial conditions to the complete solution

For $x = C\cos 20t + D\sin 20t + P\cos(10t)$, $C = x_0 - P$ and $D = v_0/20$. Thus, in meters,

$$(a) \quad x = -0.033333333\cos 20t + 0.13333333\cos(10t),$$
$$(b) \quad x = -0.13333333\cos 20t + 0.5\sin 20t + 0.13333333\cos(10t),$$
$$(c) \quad x = -0.033333333\cos 20t + 0.5\sin 20t + 0.13333333\cos(10t).$$

3. Check and interpret

At zero time, the two cosine coefficients sum to x_0; differentiating leaves $20D = v_0$. The forced amplitude is $|P|$, not the maximum of the sum. Both frequencies remain because there is no damping. The homogeneous coefficients differ among the three cases, even though the forced part is identical.



Response calculated from the displayed equations.

Question 4 | Exact resonance with initial motion — Application

An undamped system has $m = 10$ kg, $k = 4000$ N/m, and $F(t) = 400\cos(20t)$ N. Find the total response for (a) $x_0 = 0.1$ m, $v_0 = 0$; (b) $x_0 = 0$, $v_0 = 10$ m/s; (c) $x_0 = 0.1$ m, $v_0 = 10$ m/s. Interpret the plotted response.

Solution 4

1. Recognize resonance

$\omega_n = \sqrt{4000/10} = 20$ rad/s equals the forcing frequency. A trial $x_p = at\sin 20t$ gives $2m\omega_n a = F_0$, hence $a = 1$ m/s.

2. Enforce the initial conditions

This particular solution has zero initial displacement and velocity, so

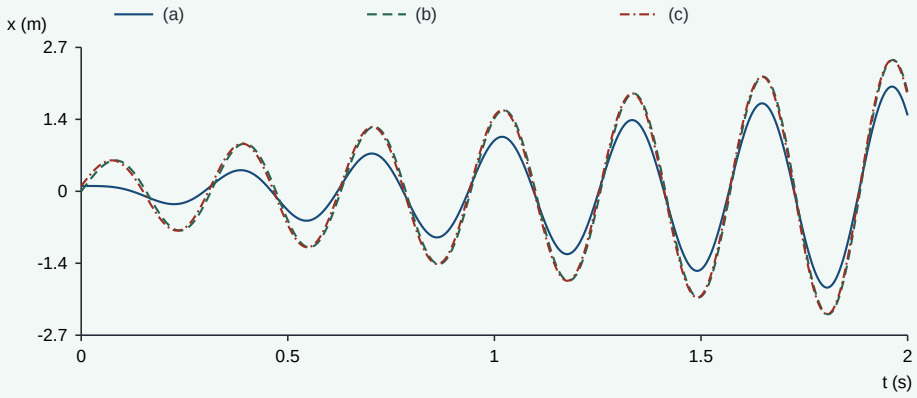
$$x = x_0\cos 20t + \frac{v_0}{20}\sin 20t + t\sin 20t.$$

The three results, in meters, are

$$(a)\quad x = 0.1\cos 20t + t\sin 20t,$$
$$(b)\quad x = (0.5 + t)\sin 20t,$$
$$(c)\quad x = 0.1\cos 20t + (0.5 + t)\sin 20t.$$

3. Interpret growth

The response is finite at every finite time but its envelope grows without bound in this ideal undamped linear model. It has no finite steady harmonic amplitude. Real damping or nonlinear limits eventually matter.



Response calculated from the displayed equations.

Question 5 | Near resonance and beating — Challenge

An undamped system has $m = 10$ kg, $k = 4000$ N/m, and $F(t) = 400\cos(20.1t)$ N. Find the total response for (a) $x_0 = 0.1$ m, $v_0 = 0$; (b) $x_0 = 0$, $v_0 = 10$ m/s; (c) $x_0 = 0.1$ m, $v_0 = 10$ m/s. Interpret the plotted response.

Solution 5

1. Compute the frequency and signed forced coefficient

$$\omega_n = 20 \text{ rad/s}, \qquad P = \frac{400}{4000 - 10(20.1)^2} = -9.9750623 \text{ m}.$$

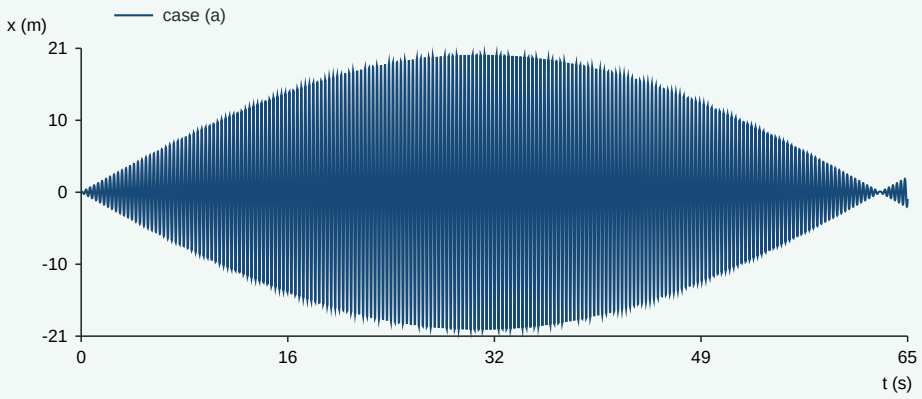
2. Apply the initial conditions to the complete solution

For $x = C\cos 20t + D\sin 20t + P\cos(20.1t)$, $C = x_0 - P$ and $D = v_0/20$. Thus, in meters,

$$\begin{aligned}
(a) \quad & x = 10.075062\cos 20t - 9.9750623\cos(20.1t), \\
(b) \quad & x = 9.9750623\cos 20t + 0.5\sin 20t - 9.9750623\cos(20.1t), \\
(c) \quad & x = 10.075062\cos 20t + 0.5\sin 20t - 9.9750623\cos(20.1t).
\end{aligned}$$

3. Check and interpret

At zero time, the two cosine coefficients sum to x_0; differentiating leaves $20D = v_0$. The forced amplitude is $|P|$, not the maximum of the sum. The close frequencies give a slow modulation with $T_b = 2\pi/|20.1 - 20| = 62.832$ s. The large predicted excursions emphasize the limits of an ideal linear undamped model. The plot shows case (a); cases (b) and (c) follow the equations above.



Response calculated from the displayed equations.

Question 6 | Above-resonance response — Application

An undamped system has $m = 10$ kg, $k = 4000$ N/m, and $F(t) = 400\cos(30t)$ N. Find the total response for (a) $x_0 = 0.1$ m, $v_0 = 0$; (b) $x_0 = 0$, $v_0 = 10$ m/s; (c) $x_0 = 0.1$ m, $v_0 = 10$ m/s. Interpret the plotted response.

Solution 6

1. Compute the frequency and signed forced coefficient

$$\omega_n = 20 \text{ rad/s}, \qquad P = \frac{400}{4000 - 10(30)^2} = -0.08 \text{ m}.$$

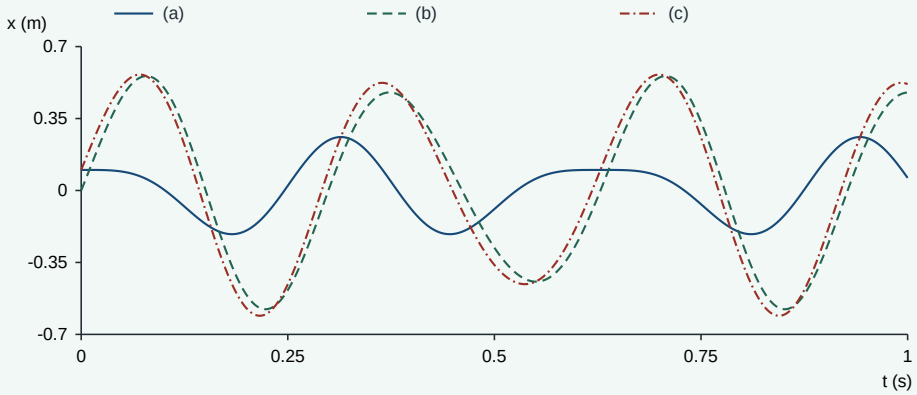
2. Apply the initial conditions to the complete solution

For $x = C\cos 20t + D\sin 20t + P\cos(30t)$, $C = x_0 - P$ and $D = v_0/20$. Thus, in meters,

$$\begin{aligned}
(a) \quad & x = 0.18\cos 20t - 0.08\cos(30t), \\
(b) \quad & x = 0.08\cos 20t + 0.5\sin 20t - 0.08\cos(30t), \\
(c) \quad & x = 0.18\cos 20t + 0.5\sin 20t - 0.08\cos(30t).
\end{aligned}$$

3. Check and interpret

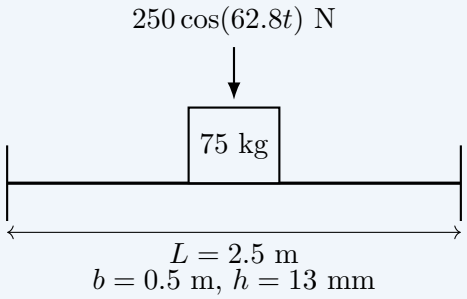
At zero time, the two cosine coefficients sum to x_0; differentiating leaves $20D = v_0$. The forced amplitude is $|P|$, not the maximum of the sum. Here $P < 0$: the forced component has a phase lag of π. The total response also contains the undamped natural-frequency component, so its peaks are not simply $|P|$.



Response calculated from the displayed equations.

Question 7 | Pump on a fixed–fixed plate — Application

A 75-kg reciprocating pump sits at midspan of a steel plate modeled as a fixed–fixed beam: $L = 2.5$ m, $b = 0.5$ m, $h = 13$ mm, $E = 200$ GPa. Its prescribed vertical force is $250\cos(62.8t)$ N. Neglect beam mass and damping. Find the harmonic displacement amplitude and phase, excluding the homogeneous response.



Solution 7

1. Find the bending stiffness

$$I = \frac{bh^3}{12} = \frac{0.5(0.013)^3}{12} = 9.15417 \times 10^{-8}\ \text{m}^4.$$

For a central load on a fixed–fixed beam, $\delta = FL^3/(192EI)$, so

$$k = \frac{192EI}{L^3} = 224972.8\ \text{N/m}.$$

2. Evaluate the dynamic stiffness

$\omega_n = \sqrt{k/m} = 54.769$ rad/s, which is below 62.8 rad/s. Thus

$$P = \frac{250}{224972.8 - 75(62.8)^2} = -0.00353032\ \text{m}.$$

3. Report amplitude and phase separately

$$\boxed{X = 3.5303\ \text{mm}, \quad \phi = \pi, \quad x_p = 0.00353032\cos(62.8t - \pi)\ \text{m}.}$$

The negative coefficient represents phase reversal, not a negative amplitude. Keeping the unrounded section moment avoids an unnecessary error in the response near resonance.

Question 8 | Identify mass from a beat-envelope measurement **Challenge**

An undamped spring–mass system has $k = 5000$ N/m and is started from rest by $30\cos(2\pi\, 20t)$ N. The maximum magnitude of the slow beat envelope is 0.2 m. Determine all positive masses consistent with this measurement. Explain what additional observation would distinguish them.

Solution 8

1. Use the release-from-rest response

$$x = P(\cos\omega t - \cos\omega_n t), \qquad P = \frac{30}{5000 - m\omega^2}, \quad \omega = 40\pi.$$

The identity $\cos a - \cos b = -2\sin[(a+b)/2]\sin[(a-b)/2]$ gives a maximum envelope magnitude $2|P|$. Thus $|P| = 0.1$ m.

2. Retain both signs of the dynamic stiffness

$$|5000 - m(40\pi)^2| = \frac{30}{0.1} = 300.$$

Therefore

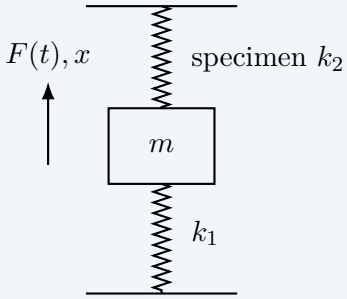
$$\boxed{m_1 = \frac{4700}{(40\pi)^2} = 0.29763 \text{ kg}, \quad m_2 = \frac{5300}{(40\pi)^2} = 0.33563 \text{ kg}.}$$

3. Resolve the ambiguity

For m_1, $\omega_n > \omega$ and $P > 0$; for m_2, $\omega_n < \omega$ and $P < 0$. Measuring the free natural frequency or the forced component's phase identifies the branch. A measured forced-component amplitude of 0.2 m would instead imply $|k - m\omega^2| = 150$ N/m. A beat-envelope maximum is also not necessarily a sampled displacement peak at exactly the same instant.

Question 9 | Electromagnetic fatigue tester — Challenge

The armature of a fatigue tester has mass $m = 20$ kg and moves between two grounded elastic elements: $k_1 = 1.79 \times 10^6$ N/m and specimen stiffness $k_2 = 135 \times 10^6$ N/m. A force $F_0 \cos \omega t$ acts on it. Neglect damping. Find all frequencies at which the specimen stress amplitude is twice the stress produced by applying F_0 directly to that specimen.



Solution 9

1. Model the armature motion

Both elements resist the same displacement, so $K = k_1 + k_2 = 136.79 \times 10^6$ N/m. The forced amplitude is $X = F_0/|K - m\omega^2|$.

2. Translate the stress requirement into a force requirement

For specimen area A_s, stress amplitude is $k_2 X/A_s$, whereas direct loading gives F_0/A_s. Thus $k_2 X = 2F_0$, or

$$|K - m\omega^2| = k_2/2 = 67.5 \times 10^6 \text{ N/m}.$$

3. Solve both frequency branches

$$\omega^2 = \frac{136.79 \times 10^6 \mp 67.5 \times 10^6}{20}, \qquad \boxed{f = 296.24 \text{ Hz or } 508.66 \text{ Hz}.}$$

The lower-frequency branch is in phase with the applied force; the upper is opposite in phase. Both satisfy the amplitude requirement. Frequency selection would also need practical actuator and specimen constraints.

Question 10 | Pressure-driven spring actuator **Application**

A diaphragm of area $A = 0.065$ m^2 drives a valve and rod of mass 12 kg against a spring of stiffness 75×10^3 N/m and mass 8 kg. Pressure fluctuates as $p(t) = 65\sin(8t)$ kPa. Use effective spring mass $m_s/3$, neglect damping and diaphragm inertia, and find the harmonic valve response. State whether the total response is determined without initial conditions.

Solution 10

1. Convert pressure into applied force

$$F(t) = Ap(t) = (0.065)(65000)\sin 8t = 4225\sin 8t \text{ N}.$$

2. Use the given one-coordinate spring-mass approximation

$$m_e = 12 + 8/3 = 14.6667 \text{ kg}, \qquad m_e\ddot{x} + 75000x = 4225\sin 8t.$$

3. Find the forced response
With $x_p = P\sin 8t$, $(75000 - 64m_e)P = 4225$, giving

$$\boxed{x_p = 0.0570473\sin 8t \text{ m}.}$$

The total response is $x = x_p + C\cos\omega_n t + D\sin\omega_n t$, where $\omega_n = \sqrt{75000/14.6667} = 71.51$ rad/s. The constants cannot be fixed without initial data. No free-vibration spring-mass derivation is required here; its effective mass is supplied and used to predict forced motion.

Question 11 | Damped forced response in sine and cosine form **Foundation**

For $m\ddot{x} + c\dot{x} + kx = F_0 \cos \omega t$ with $0 < \zeta < 1$, derive $x_p = A \cos \omega t + B \sin \omega t$, then the complete response for arbitrary x_0, v_0. Explain how to obtain a nonnegative amplitude and the correct phase quadrant.

Solution 11

1. Match the cosine and sine coefficients
Let $d = k - m\omega^2$. Substitution gives

$$dA + c\omega B = F_0, \qquad dB - c\omega A = 0.$$

Solving these two simultaneous equations,

$$A = \frac{F_0 d}{d^2 + (c\omega)^2}, \qquad B = \frac{F_0 c\omega}{d^2 + (c\omega)^2}.$$

2. Recover amplitude and phase
Since $X \cos(\omega t - \phi) = X \cos \phi \cos \omega t + X \sin \phi \sin \omega t$,

$$X = \sqrt{A^2 + B^2}, \quad \phi = \operatorname{atan2}(B, A).$$

For positive c, ω, the phase is between 0 and π. A plain inverse tangent can lose the second quadrant.

3. Impose the initial conditions on the total response
Put $a = c/(2m)$ and $\omega_d = \sqrt{k/m - a^2}$. Then

$$x = e^{-at}(C \cos \omega_d t + D \sin \omega_d t) + A \cos \omega t + B \sin \omega t.$$

At zero time $C + A = x_0$ and $-aC + \omega_d D + \omega B = v_0$, giving

$$\boxed{C = x_0 - A, \qquad D = \frac{v_0 - \omega B + aC}{\omega_d}.}$$

The factor e^{-at} must multiply both homogeneous terms. The particular solution alone generally fails the initial conditions.

Question 12 | Complete response of the lecture example **Application**

Find the steady and total responses for $m = 10$ kg, $c = 20$ N s/m, $k = 4000$ N/m, $F(t) = 100\cos(10t)$ N, $x_0 = 0.01$ m and $v_0 = 0$ m/s. Check both initial conditions.

Solution 12

1. Calculate the dynamic stiffness and damping parameters

$$d = k - m\omega^2 = 3000 \text{ N/m}, \quad c\omega = 200 \text{ N/m}.$$

$$a = \frac{c}{2m} = 1 \text{ s}^{-1}, \qquad \omega_d = \sqrt{k/m - a^2} = 19.974984 \text{ rad/s}.$$

2. Solve for the harmonic coefficients

Using $A = F_0 d/[d^2 + (c\omega)^2]$ and $B = F_0 c\omega/[d^2 + (c\omega)^2]$,

$$A = 0.033185841 \text{ m}, \quad B = 0.0022123894 \text{ m}.$$

Thus $X = \sqrt{A^2 + B^2} = 0.033259505$ m and $\phi = \operatorname{atan2}(B, A) = 0.066568164$ rad:

$$\boxed{x_p = 0.033259505\cos(10t - 0.066568164) \text{ m}.}$$

3. Enforce displacement and velocity at zero time

$$C = x_0 - A = -0.023185841 \text{ m}, \quad D = \frac{v_0 - \omega B + aC}{\omega_d} = -0.0022683239 \text{ m}.$$

The complete response, in meters, is

$$\boxed{\begin{aligned} x(t) = e^{-1t}\big[&-0.023185841\cos(19.974984t) \\ &- 0.0022683239\sin(19.974984t)\big] \\ &+ 0.033185841\cos(10t) + 0.0022123894\sin(10t). \end{aligned}}$$

4. Check both conditions

$x(0) = C + A = 0.01$ m. Differentiating gives $\dot{x}(0) = -aC + \omega_d D + \omega B = 0$ m/s. The rounded coefficients may leave a small numerical residual; the defining expressions satisfy the conditions exactly.

Question 13 | Initial displacement under off-resonance forcing **Application**

Find the steady and total responses for $m = 10$ kg, $c = 40$ N s/m, $k = 4000$ N/m, $F(t) = 200\cos(10t)$ N, $x_0 = 0.1$ m and $v_0 = 0$ m/s. Check both initial conditions.

Solution 13

1. Calculate the dynamic stiffness and damping parameters

$$d = k - m\omega^2 = 3000 \text{ N/m}, \quad c\omega = 400 \text{ N/m}.$$

$$a = \frac{c}{2m} = 2 \text{ s}^{-1}, \qquad \omega_d = \sqrt{k/m - a^2} = 19.899749 \text{ rad/s}.$$

2. Solve for the harmonic coefficients

Using $A = F_0 d/[d^2 + (c\omega)^2]$ and $B = F_0 c\omega/[d^2 + (c\omega)^2]$,

$$A = 0.065502183 \text{ m}, \quad B = 0.0087336245 \text{ m}.$$

Thus $X = \sqrt{A^2 + B^2} = 0.06608186$ m and $\phi = \text{atan2}(B, A) = 0.13255153$ rad:

$$\boxed{x_p = 0.06608186\cos(10t - 0.13255153) \text{ m}.}$$

3. Enforce displacement and velocity at zero time

$$C = x_0 - A = 0.034497817 \text{ m}, \quad D = \frac{v_0 - \omega B + aC}{\omega_d} = -0.0009216504 \text{ m}.$$

The complete response, in meters, is

$$\boxed{\begin{aligned} x(t) = e^{-2t}\big[&0.034497817\cos(19.899749t) \\ &- 0.0009216504\sin(19.899749t)\big] \\ &+ 0.065502183\cos(10t) + 0.0087336245\sin(10t). \end{aligned}}$$

4. Check both conditions

$x(0) = C + A = 0.1$ m. Differentiating gives $\dot{x}(0) = -aC + \omega_d D + \omega B = 0$ m/s. The rounded coefficients may leave a small numerical residual; the defining expressions satisfy the conditions exactly.

Question 14 | Damped startup at the natural frequency **Application**

Find the steady and total responses for $m = 10$ kg, $c = 40$ N s/m, $k = 4000$ N/m, $F(t) = 200\cos(20t)$ N, $x_0 = 0$ m and $v_0 = 0$ m/s. Check both initial conditions.

Solution 14

1. Calculate the dynamic stiffness and damping parameters

$$d = k - m\omega^2 = 0 \text{ N/m}, \quad c\omega = 800 \text{ N/m}.$$

$$a = \frac{c}{2m} = 2 \text{ s}^{-1}, \qquad \omega_d = \sqrt{k/m - a^2} = 19.899749 \text{ rad/s}.$$

2. Solve for the harmonic coefficients

Using $A = F_0 d/[d^2 + (c\omega)^2]$ and $B = F_0 c\omega/[d^2 + (c\omega)^2]$,

$$A = 0 \text{ m}, \quad B = 0.25 \text{ m}.$$

Thus $X = \sqrt{A^2 + B^2} = 0.25$ m and $\phi = \operatorname{atan2}(B, A) = 1.5707963$ rad:

$$\boxed{x_p = 0.25\cos(20t - 1.5707963) \text{ m}.}$$

3. Enforce displacement and velocity at zero time

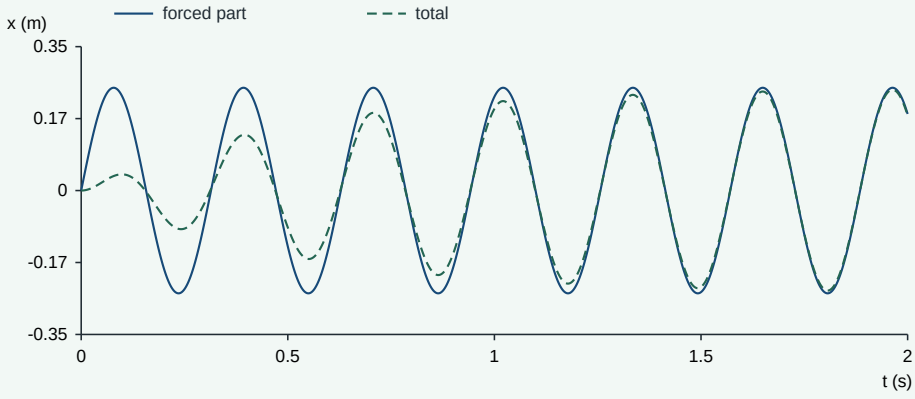
$$C = x_0 - A = 0 \text{ m}, \quad D = \frac{v_0 - \omega B + aC}{\omega_d} = -0.25125945 \text{ m}.$$

The complete response, in meters, is

$$\boxed{\begin{aligned} x(t) = e^{-2t}\big[&0\cos(19.899749t) \\ &- 0.25125945\sin(19.899749t)\big] \\ &+ 0\cos(20t) + 0.25\sin(20t). \end{aligned}}$$

4. Check both conditions

$x(0) = C + A = 0$ m. Differentiating gives $\dot{x}(0) = -aC + \omega_d D + \omega B = 0$ m/s. The rounded coefficients may leave a small numerical residual; the defining expressions satisfy the conditions exactly.



The homogeneous contribution decays; the forced part remains.

Question 15 | Above-resonance forcing with initial velocity **Application**

Find the steady and total responses for $m = 10$ kg, $c = 45$ N s/m, $k = 2500$ N/m, $F(t) = 180\cos(7\pi t)$ N, $x_0 = 0.015$ m and $v_0 = 5$ m/s. Check both initial conditions. The forcing frequency is 3.5 Hz.

Solution 15

1. Calculate the dynamic stiffness and damping parameters

$$d = k - m\omega^2 = -2336.106 \text{ N/m}, \quad c\omega = 989.6017 \text{ N/m}.$$

$$a = \frac{c}{2m} = 2.25 \text{ s}^{-1}, \qquad \omega_d = \sqrt{k/m - a^2} = 15.650479 \text{ rad/s}.$$

2. Solve for the harmonic coefficients

Using $A = F_0 d/[d^2 + (c\omega)^2]$ and $B = F_0 c\omega/[d^2 + (c\omega)^2]$,

$$A = -0.065328333 \text{ m}, \quad B = 0.02767384 \text{ m}.$$

Thus $X = \sqrt{A^2 + B^2} = 0.070948097$ m and $\phi = \operatorname{atan2}(B, A) = 2.7408986$ rad:

$$\boxed{x_p = 0.070948097\cos(7\pi t - 2.7408986) \text{ m}.}$$

3. Enforce displacement and velocity at zero time

$$C = x_0 - A = 0.080328333 \text{ m}, \quad D = \frac{v_0 - \omega B + aC}{\omega_d} = 0.2921418 \text{ m}.$$

The complete response, in meters, is

$$\boxed{\begin{aligned} x(t) = e^{-2.25t}\big[&0.080328333\cos(15.650479t) \\ &+ 0.2921418\sin(15.650479t)\big] \\ &- 0.065328333\cos(7\pi t) + 0.02767384\sin(7\pi t). \end{aligned}}$$

4. Check both conditions

$x(0) = C + A = 0.015$ m. Differentiating gives $\dot{x}(0) = -aC + \omega_d D + \omega B = 5$ m/s. The rounded coefficients may leave a small numerical residual; the defining expressions satisfy the conditions exactly.

Question 16 | A high-frequency forced response **Application**

A system has $m = 1$ kg, $c = 50$ N s/m, $k = 50000$ N/m and $F(t) = 50\cos 400t$ N. Find its steady displacement, and obtain the velocity and acceleration amplitudes from that response.

Solution 16

1. Calculate the complex dynamic stiffness

$$d = 50000 - 1(400)^2 = -110000 \text{ N/m}, \qquad c\omega = 20000 \text{ N/m}.$$

2. Calculate amplitude and phase

$$X = \frac{50}{\sqrt{110000^2 + 20000^2}} = 0.000447214 \text{ m},$$

$$\phi = \operatorname{atan2}(20000, -110000) = 2.96174 \text{ rad} = 169.70^\circ.$$

$$\boxed{x_p = 0.000447214\cos(400t - 2.96174) \text{ m}.}$$

3. Differentiate the forced motion

$$V = \omega X = 0.178885 \text{ m/s}, \qquad A_{\text{acc}} = \omega^2 X = 71.5542 \text{ m/s}^2.$$

Velocity leads displacement by 90° and acceleration by 180°. A small displacement at high frequency can still imply substantial acceleration; the numerical units should not be interchanged.

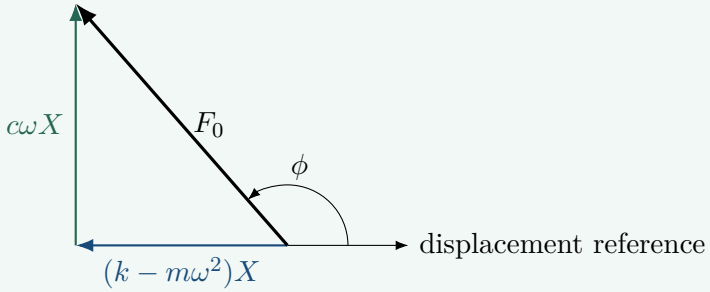
Question 17 | Graphical force-vector solution **Application**

Use a force-vector diagram to determine the steady response for $m = 150$ kg, $k = 25$ kN/m, $c = 2000$ N s/m and $F(t) = 100\cos 20t$ N. Show the correct phase quadrant.

Solution 17

1. Construct the equation-term vectors relative to displacement

Per unit displacement, stiffness points right with magnitude 25000 N/m and the acceleration term points left with magnitude $m\omega^2 = 60000$ N/m. Their resultant is -35000 N/m. Damping points upward with magnitude $c\omega = 40000$ N/m.



2. Use the triangle magnitude

$$100 = X\sqrt{35000^2 + 40000^2}, \qquad X = 0.00188144 \text{ m}.$$

3. Use the quadrant-II angle

$$\phi = \operatorname{atan2}(40000, -35000) = 2.28963 \text{ rad} = 131.19^\circ.$$

$$\boxed{x_p = 0.00188144\cos(20t - 2.28963) \text{ m}.}$$

The force leads the displacement. These are phasors of the left-hand-side terms of the motion equation, not physical restoring-force arrows on a free-body diagram.

Question 18 | Where is the displacement peak? **Foundation**

For fixed harmonic force amplitude, derive the frequency ratio and value of the maximum displacement magnification. State the damping range in which a positive-frequency peak exists. Distinguish it from ω_n and ω_d.

Solution 18

1. Minimize the squared denominator
Let $M = X/(F_0/k) = [(1-r^2)^2 + 4\zeta^2 r^2]^{-1/2}$. Set $g = (1-r^2)^2 + 4\zeta^2 r^2$. Then

$$g'(r) = 4r(r^2 + 2\zeta^2 - 1).$$

For $0 < \zeta < 1/\sqrt{2}$, the interior minimum is at $r_p = \sqrt{1-2\zeta^2}$.
2. Evaluate the gain at this frequency
Substituting $r_p^2 = 1 - 2\zeta^2$ gives $g(r_p) = 4\zeta^2(1-\zeta^2)$. Thus

$$\boxed{\omega_p = \omega_n\sqrt{1-2\zeta^2}, \quad M_{\max} = \frac{1}{2\zeta\sqrt{1-\zeta^2}}.}$$

3. Check the boundary cases
For $\zeta \geq 1/\sqrt{2}$, the largest gain is $M(0) = 1$ and there is no positive-frequency displacement peak. With no damping the resonant gain is unbounded. For $0 < \zeta < 1/\sqrt{2}$, $\omega_p < \omega_d = \omega_n\sqrt{1-\zeta^2} < \omega_n$. At $\omega = \omega_n$ the phase is $90°$ and the gain is $1/(2\zeta)$, but this is not exactly the displacement maximum.

Question 19 | Velocity and acceleration frequency responses **Foundation**

Derive the normalized velocity amplitude $V/(F_0/\sqrt{km})$ and acceleration amplitude $A_{\text{acc}}/(F_0/m)$ as functions of r and ζ. Compare their low- and high-frequency limits with the displacement response.

Solution 19

1. Use the steady harmonic derivatives

$V = \omega X$ and $A_{\text{acc}} = \omega^2 X$. Define $D = \sqrt{(1-r^2)^2 + (2\zeta r)^2}$, so $X = F_0/(kD)$ and $\omega = r\sqrt{k/m}$.

2. Normalize using the specified scales

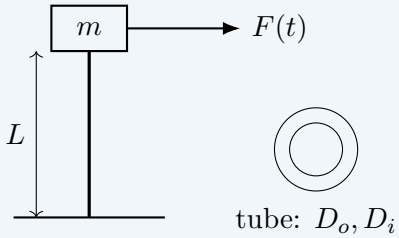
$$\boxed{\frac{V}{F_0/\sqrt{km}} = \frac{r}{D}, \qquad \frac{A_{\text{acc}}}{F_0/m} = \frac{r^2}{D}.}$$

3. Take the limits

As $r \to 0$, $D \to 1$: displacement tends to the static deflection, while velocity and acceleration gains tend to zero. As $r \to \infty$, $D \sim r^2$: displacement and velocity gains tend to zero, but acceleration gain tends to one. Thus inertia balance gives $A_{\text{acc}} \to F_0/m$. At $r = 1$, all three normalized gains equal $1/(2\zeta)$ for $\zeta > 0$, although their dimensional amplitudes and the locations of their maxima differ.

Question 20 | Design a tubular camera support — Challenge

A 2-kg camera is mounted on a vertical cantilevered aluminum tube, $L = 0.5$ m and $E = 70$ GPa. A lateral force $25\cos(75.3984t)$ N acts on the camera. Neglect tube mass and damping. Find the allowable relation between outer and inner diameters if the steady displacement amplitude must not exceed 5 mm. Is one unique pair of diameters determined?



Solution 20

1. Relate tube geometry to stiffness

$$I = \frac{\pi}{64}(D_o^4 - D_i^4), \quad k = \frac{3EI}{L^3}, \quad 0 \le D_i < D_o.$$

2. Use the amplitude constraint with an absolute value

$$\frac{25}{|k - 2(75.3984)^2|} \le 0.005 \quad \Rightarrow \quad |k - 11369.79| \ge 5000.$$

There are two admissible stiffness ranges: $0 < k \le 6369.79$ N/m or $k \ge 16369.79$ N/m.

3. Convert to section-moment ranges

$$\boxed{0 < I \le 3.79157 \times 10^{-9}\ \text{m}^4 \quad \text{or} \quad I \ge 9.74395 \times 10^{-9}\ \text{m}^4.}$$

Substitute $I = \pi(D_o^4 - D_i^4)/64$ to obtain the allowed diameter pairs. For the stiff branch,

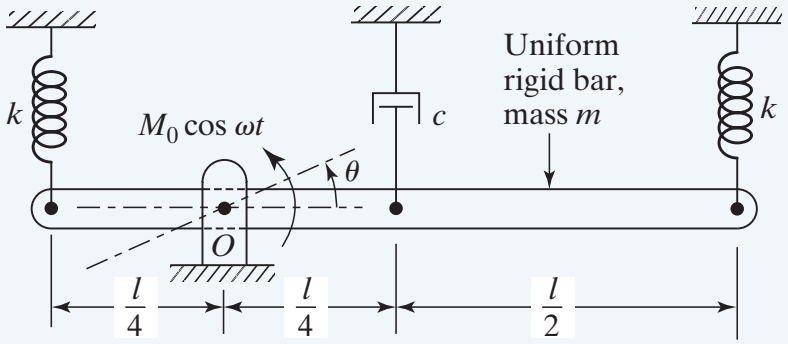
$$D_o^4 - D_i^4 \ge 1.9850 \times 10^{-7}\ \text{m}^4.$$

4. Interpret the design freedom

The amplitude requirement supplies one inequality, not two independent dimensions. Wall thickness, strength, static deflection and manufacturing restrictions would be needed for a unique choice. The soft branch is a mathematical possibility at the specified operating frequency; startup near resonance and structural limits still require assessment.

Question 21 | Harmonic torque on a spring-supported bar — Challenge

The uniform bar has $m = 10$ kg and $l = 1$ m. Its hinge is $l/4$ from the left end; two end springs have $k = 5000$ N/m each. A vertical damper $c = 1000$ N s/m acts at the midpoint. The applied torque is $100\cos\omega t$ N m, with forcing frequency 1000 cycles/min. Derive the small-rotation equation and find the steady response.



Distances are measured along the bar; rotation is about the fixed hinge.

Solution 21

1. Calculate inertia about the hinge

The center of mass is $l/4$ from the hinge, so

$$J_O = \frac{ml^2}{12} + m(l/4)^2 = \frac{7ml^2}{48} = 1.458333 \text{ kg m}^2.$$

2. Refer spring and damper forces to angular motion

An element at signed distance a moves by $a\theta$; its restoring moment coefficient is proportional to a^2. Thus

$$K_\theta = k[(l/4)^2 + (3l/4)^2] = 3125 \text{ N m/rad},$$

$$C_\theta = c(l/4)^2 = 62.5 \text{ N m s/rad}.$$

The incremental equation is $1.458333\ddot{\theta} + 62.5\dot{\theta} + 3125\theta = 100\cos\omega t$.

3. Convert frequency and find the phasor response

$\omega = 1000(2\pi/60) = 104.7198$ rad/s. Therefore

$$\Theta = \frac{100}{\sqrt{(3125 - J_O\omega^2)^2 + (62.5\omega)^2}} = 0.00692698 \text{ rad},$$

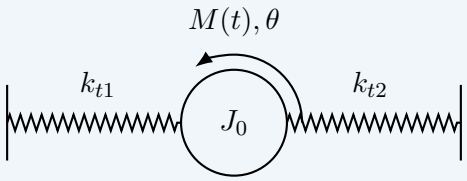
$$\phi = \text{atan2}(62.5\omega, 3125 - J_O\omega^2) = 2.67105 \text{ rad}.$$

$$\boxed{\theta_p = 0.00692698\cos(104.7198t - 2.67105) \text{ rad}.}$$

The damper lever arm is $l/4$ from the hinge, not $l/2$; using the wrong reference would multiply its angular damping by four.

Question 22 | A rotor driven by harmonic torque — Application

A rotor has $J_0 = 0.05$ kg m^2. Two shaft segments connect it to fixed ends with torsional stiffnesses 3000 and 4000 N m/rad. Equivalent torsional damping is $c_t = 2.5$ N m s/rad. Find the steady response to $M(t) = 200\cos 500t$ N m.



equivalent torsional damping c_t to ground

Solution 22

1. Combine restoring torques

Both fixed-end shafts resist the rotor's same rotation, so $K_t = 3000+4000 = 7000$ N m/rad. The motion equation is

$$0.05\ddot{\theta} + 2.5\dot{\theta} + 7000\theta = 200\cos 500t.$$

2. Calculate the complex angular response

$$\hat{\Theta} = \frac{200}{7000 - 0.05(500)^2 + i(2.5)(500)} = \frac{200}{-5500 + i1250}.$$

3. Convert to amplitude and phase

$$\Theta = \frac{200}{\sqrt{5500^2 + 1250^2}} = 0.0354594 \text{ rad},$$

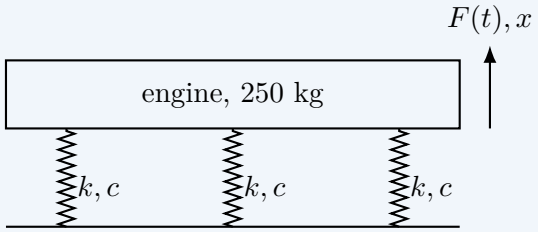
$$\phi = \operatorname{atan2}(1250, -5500) = 2.91812 \text{ rad}.$$

$$\boxed{\theta_p = 0.0354594\cos(500t - 2.91812) \text{ rad}.}$$

The response is above resonance: $\omega_n = \sqrt{7000/0.05} = 374.166$ rad/s. This is applied torque excitation, not rotating-unbalance forcing.

Question 23 | Three engine mounts: is the design unique? Challenge

A 250-kg engine is supported by three identical mounts, each of stiffness k and viscous coefficient c. Model pure vertical translation under the prescribed force $1000\sin(100\pi t)$ N. Design the mounts so the steady amplitude is less than 2.5 mm with system damping ratio $\zeta = 0.01$. Explain whether these requirements determine a unique design.



Each mount symbol represents a spring and viscous damper in parallel.

Solution 23

1. Combine the mounts

$K = 3k$, $C = 3c$ and $C = 2\zeta\sqrt{mK}$. At $\omega = 100\pi$ rad/s, let $H = m\omega^2 = 24.6740 \times 10^6$ N/m. Then

$$X = \frac{1000}{\sqrt{(K-H)^2 + 4\zeta^2 KH}}.$$

2. Check the worst value as stiffness is varied

Completing the square gives

$$(K-H)^2 + 4\zeta^2 KH = [K - H(1-2\zeta^2)]^2 + 4\zeta^2(1-\zeta^2)H^2.$$

Consequently, for the stipulated damping ratio,

$$X \le \frac{1000}{2\zeta H\sqrt{1-\zeta^2}} = 2.02653 \text{ mm} < 2.5 \text{ mm}.$$

The given operating-frequency displacement limit therefore does not select a unique stiffness: every positive K, with damping adjusted to keep $\zeta = 0.01$, meets it in this ideal model.

3. Give one checkable design

One algebraically convenient choice is $K = m\omega^2$. It gives, per mount,

$$\boxed{k = K/3 = 8.22467 \times 10^6 \text{ N/m}, \quad c = \frac{2\zeta m\omega}{3} = 523.599 \text{ N s/m}.}$$

Its response amplitude is $1000/(C\omega) = 2.02642$ mm. This is a feasible example, not an optimized isolator. Static travel, startup, support forces and practical mount properties are additional design constraints outside the stated problem.

Question 24 | Damping from two forced amplitudes — Application

With the same force amplitude in both tests, a viscously damped system has displacement amplitude 20 mm at $\omega = \omega_n$ and 10 mm at $\omega = 0.75\omega_n$. Determine ζ. Here the first test is at the undamped natural frequency, not the exact peak-displacement frequency.

Solution 24

1. Write the two amplitudes using the same static scale

Let $\delta = F_0/k$. Then $X_1 = \delta/(2\zeta)$ and

$$X_{.75} = \frac{\delta}{\sqrt{(1-0.75^2)^2 + (1.5\zeta)^2}}.$$

2. Eliminate the unknown force/stiffness scale

$$\frac{20}{10} = \frac{\sqrt{0.4375^2 + 2.25\zeta^2}}{2\zeta}.$$

Squaring gives $16\zeta^2 = 0.19140625 + 2.25\zeta^2$. Therefore

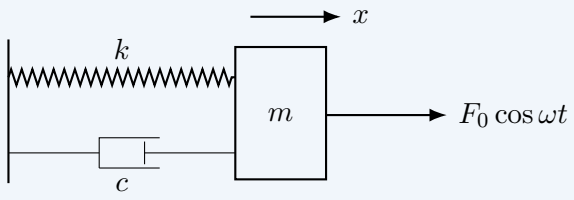
$$\boxed{\zeta = \sqrt{0.19140625/13.75} = 0.117985.}$$

3. Verify the readings

$\delta = 2\zeta X_1 = 4.71940$ mm. The second denominator is $4\zeta = 0.471940$, so $X_{.75} = 10$ mm. This inference depends on constant force amplitude and the stated frequency reference. Interpreting the word resonance as the exact peak instead would require a different pair of equations.

Question 25 | Identify an air-compressor foundation — Challenge

A 100-kg compressor is supported by an elastic, viscously damped foundation. During a constant-amplitude 100-N force sweep, the maximum steady displacement is 5 mm at 300 cycles/min. Determine the foundation stiffness and damping coefficient using the exact displacement-peak relations.



Solution 25

1. Express the measured peak frequency correctly

$$\omega_p = 300(2\pi/60) = 10\pi \text{ rad/s}.$$

Set $a = c/(2m)$ and $\omega_n^2 = k/m$. The exact peak relation gives $\omega_p^2 = \omega_n^2 - 2a^2$.

2. Use the measured peak amplitude

At the peak, $k - m\omega_p^2 = 2ma^2$ and $c\omega_p = 2ma\omega_p$. Hence

$$X_p = \frac{F_0}{2ma\sqrt{a^2 + \omega_p^2}}.$$

Since $\omega_n^4 = (\omega_p^2 + 2a^2)^2 = \omega_p^4 + 4a^2(a^2 + \omega_p^2)$,

$$\omega_n^2 = \sqrt{\omega_p^4 + \left(\frac{F_0}{mX_p}\right)^2} = 1007.02081 \text{ s}^{-2}.$$

3. Recover stiffness and damping

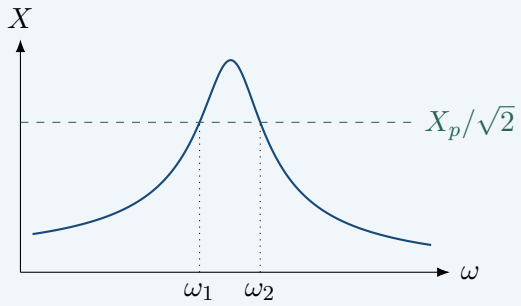
$$k = m\omega_n^2 = \boxed{100702.1 \text{ N/m}},$$

$$a = \sqrt{(\omega_n^2 - \omega_p^2)/2} = 3.16705 \text{ s}^{-1}, \quad c = 2ma = \boxed{633.409 \text{ N s/m}}.$$

$\zeta = a/\omega_n = 0.09980 < 1/\sqrt{2}$, so the assumed positive-frequency peak exists. Substitution in the amplitude equation returns 0.005 m. Equating the measured peak frequency directly to ω_n would introduce an approximation.

Question 26 | Quality factor and half-power bandwidth — Foundation

For a lightly damped, isolated SDOF resonance under a constant-force sweep, derive the bandwidth estimate of damping and its relation to quality factor Q. Explain why the amplitude threshold is the peak divided by $\sqrt{2}$, and show the equivalent formula when frequencies are measured in Hz.



Solution 26

1. Approximate the response near a narrow peak

For $r \approx 1$, $1 - r^2 \approx -2(r-1)$ and $2\zeta r \approx 2\zeta$. Thus

$$M \approx \frac{1}{2\sqrt{(r-1)^2 + \zeta^2}}, \qquad M_p \approx \frac{1}{2\zeta} = Q.$$

2. Locate the two threshold crossings

Set $M = M_p/\sqrt{2}$. Squaring and rearranging gives $(r-1)^2 = \zeta^2$, so $r_1 \approx 1 - \zeta$ and $r_2 \approx 1 + \zeta$.

$$\boxed{\Delta\omega = \omega_2 - \omega_1 \approx 2\zeta\omega_n, \qquad \zeta \approx \frac{\Delta\omega}{2\omega_n}, \quad Q \approx \frac{\omega_n}{\Delta\omega}.}$$

Also $\omega_1 + \omega_2 \approx 2\omega_n$, giving $\zeta \approx (\omega_2 - \omega_1)/(\omega_2 + \omega_1)$.

3. Interpret the threshold and measurement conditions

The squared amplitude at $X_p/\sqrt{2}$ is half its peak value; dividing amplitude by two would quarter it. Actual mean viscous power is $c\omega^2 X^2/2$, so the displacement half-power interpretation is a narrow-band approximation. Keep force amplitude constant, or use displacement divided by measured force; allow the response to settle at each frequency.

4. Use consistent units

Since $\omega = 2\pi f$, the factors cancel: $\boxed{\zeta \approx (f_2 - f_1)/(2f_n),\ Q \approx f_n/(f_2 - f_1)}$. For light damping $f_n \approx f_p$. Do not mix Hz and rad/s in one ratio. The plot is an illustrative normalized curve at $\zeta = 0.05$, not an experimental dataset.

Question 27 | A high-Q electromechanical resonator **Application**

An electromechanical resonator must have $f_n = 1000$ Hz and $Q = 1200$. Estimate its damping ratio, half-power bandwidth and two half-power frequencies. Indicate the accuracy assumption.

Solution 27

1. Infer the damping ratio

For light viscous damping, $Q \approx 1/(2\zeta)$. Therefore

$$\boxed{\zeta \approx \frac{1}{2400} = 4.16667 \times 10^{-4}.}$$

This is very small, so the narrow-band approximation is appropriate.

2. Find the bandwidth in Hz and rad/s

$$\boxed{\Delta f \approx \frac{f_n}{Q} = 0.833333 \text{ Hz}}, \qquad \Delta\omega = 2\pi\Delta f \approx 5.23599 \text{ rad/s}.$$

3. Locate the two frequencies to leading order

The narrow band is approximately symmetric about f_n, hence

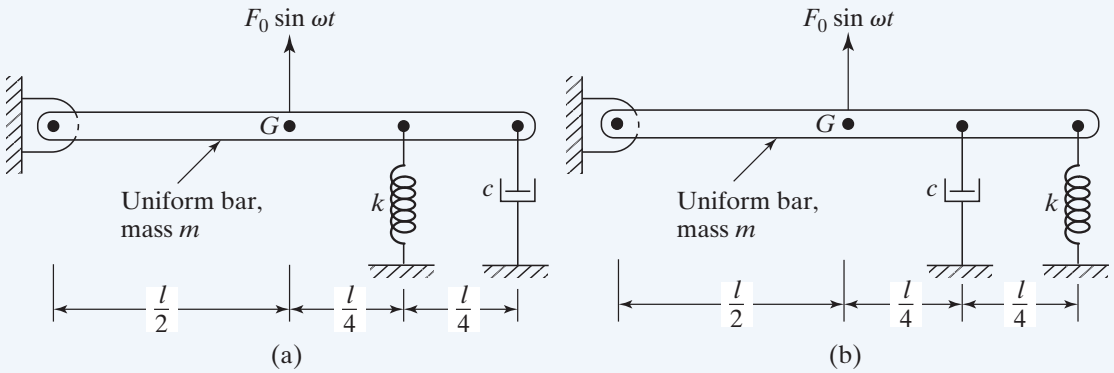
$$\boxed{f_1 \approx 1000 - 0.416667 = 999.583333 \text{ Hz},}$$

$$\boxed{f_2 \approx 1000 + 0.416667 = 1000.416667 \text{ Hz}.}$$

The exact displacement peak and exact threshold crossings are slightly shifted; the displayed values are first-order estimates, not exact symmetric roots. A frequency sweep must have resolution substantially finer than this bandwidth to estimate damping well.

Question 28 | Which bar-support arrangement gives less motion? Challenge

A uniform slender bar is hinged at its left end and driven at its midpoint by $F_0 \sin \omega t$. In arrangement (a), the spring k is at $3l/4$ and damper c at l. In arrangement (b), those positions are exchanged. Compare the steady midpoint amplitudes and determine whether one arrangement is better at every frequency.



Both models have the same bar, spring, damper and forcing.

Solution 28

1. Refer each element to the hinge coordinate

For small angle θ, $J = ml^2/3$ and applied torque amplitude $T_0 = F_0 l/2$. The two arrangements have

$$K_a = \frac{9}{16}kl^2, \quad C_a = cl^2, \qquad K_b = kl^2, \quad C_b = \frac{9}{16}cl^2.$$

2. Write the midpoint amplitudes

Because midpoint displacement is $(l/2)\theta$,

$$X_j = \frac{F_0 l^2/4}{\sqrt{(K_j - J\omega^2)^2 + (C_j\omega)^2}}, \qquad j = a, b.$$

The larger denominator gives the smaller response.

3. Compare the squared denominators

Let $K = kl^2$, $C = cl^2$, $\alpha = 9/16$, and $H = J\omega^2$. Then

$$D_a^2 - D_b^2 = (1-\alpha)\Big[2KH - (1+\alpha)K^2 + (1+\alpha)C^2\omega^2\Big].$$

Since $1 - \alpha > 0$, arrangement (a) gives less motion when

$$\boxed{\omega^2 > \frac{(1+\alpha)K^2}{2KJ + (1+\alpha)C^2}.}$$

Below this crossover, (b) gives less motion; at equality their amplitudes coincide. In particular (b) is stiffer and better in the static limit, while (a) places the damper at a larger lever arm. There is no frequency-independent winner for all operating conditions.

Question 29 | When the light-damping estimate is not enough — Challenge

A force sweep gives a peak displacement of 5 mm. The static displacement under the same force amplitude is 2.5 mm, and $f_n = 5$ Hz. Find the damping ratio using the exact peak formula, then find the frequencies where displacement amplitude is the actual peak divided by $\sqrt{2}$. Compare with a light-damping estimate.

Solution 29

1. Infer damping from the actual peak

$M_p = 5/2.5 = 2$. Thus $2 = [2\zeta\sqrt{1-\zeta^2}]^{-1}$. Setting $u = \zeta^2$ gives

$$u(1-u) = \frac{1}{16}, \quad u = \frac{1 \pm \sqrt{3/4}}{2}.$$

The peak formula requires $u < 1/2$; only the smaller root is valid:

$$\boxed{\zeta = 0.258819.}$$

2. Find the actual peak frequency

$$f_p = f_n\sqrt{1-2\zeta^2} = 4.65302 \text{ Hz}.$$

3. Solve for the two amplitude-threshold frequencies

At $M = M_p/\sqrt{2}$, the squared denominator is twice its minimum:

$$(1-r^2)^2 + 4ur^2 = 8u(1-u).$$

Solving the quadratic in r^2 gives

$$r_{1,2}^2 = 1 - 2u \mp 2\sqrt{u(1-u)}.$$

Here $1-2u = 0.8660254$ and $2\sqrt{u(1-u)} = 0.5$, hence

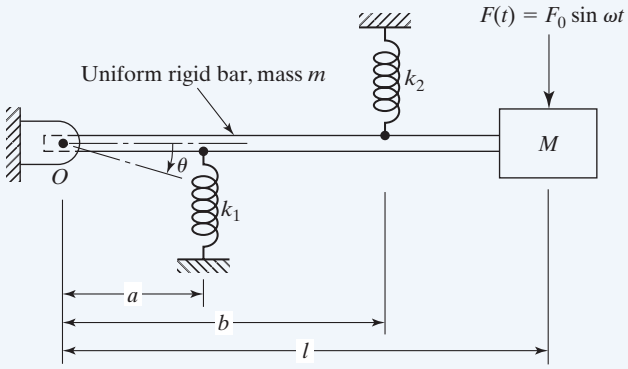
$$\boxed{f_1 = 3.02500 \text{ Hz}, \quad f_2 = 5.84385 \text{ Hz}.}$$

4. Assess the simple estimates

Using $Q \approx M_p = 2$ gives $\zeta \approx 0.25$. Using the computed bandwidth gives $(f_2 - f_1)/(2f_n) = 0.281885$. Neither is exact at this damping level. The threshold is $5/\sqrt{2} = 3.53553$ mm. These are displacement half-power points; their physical power interpretation is approximate outside a narrow resonance.

Question 30 | A forced rigid bar with a concentrated tip mass — Challenge

A uniform rigid bar of length $l = 1$ m and mass $m = 10$ kg is hinged at O and carries a tip mass $M = 50$ kg. Vertical springs $k_1 = k_2 = 5000$ N/m act at $a = 0.25$ m and $b = 0.5$ m from O. The tip is driven downward by $500 \sin \omega t$ N, at 1000 cycles/min. Neglect damping. Derive the equation about static equilibrium and find the harmonic angle and tip displacement.



Small clockwise rotation is positive; both springs supply restoring moments.

Solution 30

1. Include both distributed and concentrated inertia

$$J_O = \frac{ml^2}{3} + Ml^2 = \frac{10}{3} + 50 = 53.3333 \text{ kg m}^2.$$

2. Form the restoring and applied moments

Spring deflections are $a\theta$ and $b\theta$. Therefore

$$K_\theta = k_1 a^2 + k_2 b^2 = 5000(0.25^2 + 0.5^2) = 1562.5 \text{ N m/rad}.$$

The force moment is $F_0 l \sin \omega t$. Constant gravitational moments are balanced by the static preload; they do not appear in the incremental linear equation:

$$\boxed{53.3333\ddot{\theta} + 1562.5\theta = 500 \sin \omega t.}$$

3. Calculate the forced angular coefficient

$\omega = 1000(2\pi/60) = 104.7198$ rad/s. A trial $\theta_p = P_\theta \sin \omega t$ gives

$$P_\theta = \frac{500}{1562.5 - 53.3333(104.7198)^2} = -0.000857188 \text{ rad}.$$

$$\boxed{\theta_p = -0.000857188 \sin(104.7198t) \text{ rad}, \quad X_{\text{tip}} = l|P_\theta| = 0.857188 \text{ mm}.}$$

The forced motion is opposite in phase to the applied force. With no damping, any homogeneous motion caused by initial conditions persists; the requested harmonic response does not specify it.